Recitation Assignment #6

October 30, 2024

Today's assignment will give you a chance to get an "intuition" (if such a thing is possible) for relativistic kinematics. You will model a rocket-ship trip to α-Centauri (the nearest star to our own), about 4.3 light years from earth. Remember, a light year is just the distance that light travels in one year.

New Concepts:

Physical:

- Time Dilation
- Relativistic Energy

For this assignment:

1. To illustrate the trip, but the earth at one side of the window (the origin), and Alpha-Centauri at the other. I strongly urge you to use years and light-years as your units.
2. Make the spaceship have a mass of 1000kg, and apply a constant force of 9800N. To the inhabitants of the ship, this would feel like earth normal gravity pulling them toward the "back" of the ship. You should be *very* careful with this. The speed of the ship is not given by the non-relativistic limit. Rather:

   $$\Delta\vec{p} = \vec{F}\Delta t$$

   As a side note:

   $$9.8\frac{m}{s^2} = 1.03\ ly/yr^2$$

   Since you're not using force to directly change the velocity, you'll instead need to keep track of the total momentum, $\vec{p}$, in order to convert it to the velocity. We'll discuss in class how we get this equation, but the conversion (starting with $\boldsymbol{p = mv\gamma}$) is:

   $$v = \frac{(p/m)}{\sqrt{1 + (\frac{p}{mc})^2}}$$

3. Halfway through the trip, reverse thrusters and decelerate with force -F (flip the direction of the force). This will allow you to stop when you reach α-Centauri.
4. You will get a maximum of 80% on this if you model the ship as just a box or a sphere. Creativity in modeling your ship counts. You may find the "compound" object in vpython to be helpful. Be sure to look at the documentation to see how this works.
5. Besides the graphics, your program should compute how long, in years, such a trip takes. The time should be as seen from earth. Make sure you include this number in your email to Kendall.
6. We have not yet seen this, but there is another important concept in Special relativity: Time runs slow for bodies in motion. In particular, the amount of time that passes on the ship is:

$$\Delta t_{ship} = \frac{\Delta t_{earth}}{\gamma}$$

   This is known as "time dilation." Please compute how long the entire trip takes *according to the pilot in the ship.* Your answer should be years and should also be included in your email to Kendall.

   For reference, we introduced the relativistic constant γ early in this course. It will increase towards infinity as the speed of an object approaches the speed of light. You will compute this using the magnitude of your ship's velocity:

$$\gamma = \frac{1}{\sqrt{1 - (\frac{v}{c})^2}}$$

7. **For extra credit:** Bodies in motion also appear foreshortened by a factor of γ:

$$\Delta L = \frac{\Delta L_0}{\gamma}$$

   Figure out how to visibly give your ship an appropriate length contraction at every point on the voyage (as seen by earth, of course

As always, please make sure you test it and submit your final program to Kendall and Dr. Cruz via email.